INCH-POUND

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A-A-59419
22 December 2003

COMMERCIAL ITEM DESCRIPTION

CALIBRATOR, THERMOMETER

The General Services Administration has authorized the use of the Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This Commercial Item Description (CID), describes a free standing thermometer calibrator (TC) that is used as a means to calibrate temperature devices.

2. SALIENT CHARACTERISTICS

2.1 Performance. The TC shall have two independent bath systems, one for hot temperatures and the other for cold temperatures. Each bath system shall include a tank, temperature controller, test well, and pump. Each bath system shall have a system incorporated to heat and cool the bath. Each bath system shall be capable of maintaining the temperature that is set by the temperature controller without deviating by more than 0.1 degrees Celsius. The hot bath shall be capable of stabilizing at a set point temperature ranging between 60 to 300 degrees Celsius (140 to 572 degrees Fahrenheit). The cold bath shall be capable of stabilizing at a set point temperature ranging between -100 to 70 degrees Celsius (-148 to -158 degrees Fahrenheit). Each bath shall have a digital readout capable of displaying the set point temperature and the temperature of the tank. The TC shall be accurate in achieving the temperatures for which it is set (within 5 degrees Celsius). The two baths shall provide an accurate, stable, uniform temperature medium (not more than 0.1 degree Celsius temperature variance at any given time at any location in the tank). The two baths shall be capable of reaching and stabilizing at the temperature for which it is set in a reasonable amount of time (not more one hour and thirty minutes). Both baths shall have a heating system. The cold bath shall be incorporate a refrigeration system. The TC shall have an indicator that lets the user know when temperature stabilization for a given set point has been reached for the baths.

2.2 Controls. All controls on the TC shall be clearly marked as to identify their functionality. No calculations shall be needed to set the temperature of the baths (i.e. the user shall not have to input things such as room temperature and atmospheric pressure).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data that may improve this document should be sent to: WR-ALC/LEEE, 295 Byron Street, Robins AFB, GA 31098-1611

FSC 6685

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2.3 Electrical Power Sources. The equipment shall operate from either of the following power sources: 110, 120, 220, or 240 volts ac (± 10%) at line frequencies of 48 Hz to 66 Hz.

2.4 Fluids. The manufacturer shall recommend the fluids that should be used in the tanks. The TC shall be shipped with no fluid in the tanks. The manufacturer is encouraged to avoid the use of ozone depleting substances (ODS) in the refrigeration system (heating system if applicable).

2.5 Dimensions and Weight. The test wells shall have a depth between 11 and 15 inches, and a diameter between 2.5 and 3.5 inches. The overall size of the system shall be consistent with current commercial characteristics. The weight of the system should not exceed 600 pounds.

2.6 Safety. The equipment shall be Underwriters Laboratories (UL) listed and approved and/or shall comply with the safety requirements of MIL-PRF-28800 for the classification stated herein.

2.7 Manuals. The TC shall be delivered with operator, maintenance, and illustrated parts manuals. Format and quantity shall be as specified in the contract or order.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. PRODUCT CONFORMANCE PROVISIONS

4.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial marketplace. Small businesses that cannot offer products with established commercial market acceptability may compete by submission of bid samples. The government reserves the right to require proof of such conformance.

4.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of ASTM-SI 10, and all other requirements of this CID are met.

4.3 Exceeded Metric Products. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

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6. NOTES.

6.1 Sources of documents.

6.1.1 Government Documents. Military Specifications, Standards, and Handbooks referenced herein may be obtained from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

6.1.2 Underwriters Laboratories (UL).Copies of UL standards can be obtained from UL Corporate Headquarters, 333 Pfingsten Road, Northbrook IL 60062-2096. Phone: 1-847-272-8800. Fax 1-847-407-1395.

6.1.3 Code of Federal Regulations (CRF). Copies of the CFR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

6.1.4 ASTM standards. Copies of ASTM standards can be obtained from ASTM, 100 Bar Harbor Drive, West Conshohocken, PA 19428-2959. Phone 1-610-832-9500. Fax 1-610-832-9555.

6.2 Known NSNs. NSN 6685-01-224-4282.

MILITARY INTEREST

Custodian:
Air Force – 99

Preparing activity:
Air Force - 84

Agent:
Air Force - 99

(Project 6685-0071)

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